The BIND Lightcone Suite V: How Anisotropic Is the Baryonic Suppression of Weak Lensing?

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Abstract

Baryonification methods (BCMs) correct dark-matter-only (DMO) simulations for baryonic physics by displacing each halo's mass *radially*, which by construction discards any angular structure in the resulting correction. Whether that discarded structure matters for weak lensing (WL) has not previously been measured at the field level, because doing so requires a generative model that paints hydrodynamic-quality baryon fields onto DMO halos one-by-one so that a literal spherical counterfactual can be built and differenced against the full field. We use BIND, a conditional flow-matching emulator trained on the CAMELS-IllustrisTNG suite, to construct exactly this counterfactual: a mass-exact, per-halo radial "BCM-warp" of the painted field that conserves each halo's monopole mass profile while removing only its angular ($m \geq 2$) structure. On the fiducial-feedback lightcone ($z_{\rm s} = 1$), the fraction of the convergence power-spectrum suppression that survives this symmetrization is scale-dependent: $f_{\rm aniso} \approx 0.45$ at $\ell = 3 \times 10^3$, falling to 0.24 at $\ell = 10^4$ and to ≈ 0 by $\ell = 2 \times 10^4$. This corrects an earlier, cruder estimate that forced the correction *itself* to be azimuthally symmetric and thereby credited a real BCM with zero triaxiality response; that estimate returns a roughly scale-flat $f_{\rm mono} \approx 0.5$, which we show is dominated at small scales by halo triaxiality a real BCM already reproduces, not by missed feedback physics. Even so, the corrected signal is non-negligible in amplitude at $\ell \lesssim 10^4$ [TODO: a formal bootstrap/significance test for the corrected `bcm_warp`-control $f_{\rm aniso}$ has not yet been computed in our source material; the only significance number available ($z = 52.3\sigma$, §3.2) belongs to the superseded `mono` control, not this one] and its existence at any scale means spherically-symmetric BCMs must over-eject gas radially to fit the two-point function, biasing the recovered gas profile in a way that plausibly explains why such models fail non-Gaussian statistics. We identify AGN kinetic-mode and stellar-wind feedback parameters as the strongest controls of the (upper-bound) angular signal from 60 one-at-a-time TNG feedback variations. This is a first, preliminary field-level measurement at fixed cosmology and fiducial feedback; we flag explicitly which parts of the argument still rest on the superseded control.

1 Introduction

Hydrodynamical simulations are too expensive to run at the volumes and parameter-space coverage required for stage-IV weak-lensing (WL) cosmology, so baryonic effects on the matter power spectrum are instead modeled by *baryonification* – post-hoc, analytic corrections applied to cheap DMO simulations (Schneider and Teyssier, 2015; Aricò et al., 2020; Mead et al., 2021). Every widely used baryonification model shares one structural assumption: gas and stars are redistributed *radially* within each halo, $r \to r'(r)$, so that the correction to the DMO density field is by construction azimuthally symmetric around the halo center. This assumption is convenient – it reduces the correction to a one-dimensional profile per halo, calibrated against the spherically averaged gas fraction and pressure profiles of hydrodynamic simulations – but it is also, on its face, unphysical: feedback-driven gas ejection follows the halo potential and large-scale environment, neither of which is spherical, and the resulting anisotropy of real hydrodynamic gas maps is well documented.[TODO: find and cite a reference for angular/quadrupole structure in hydrodynamic halo gas maps; van Daalen et al. (2020) (arXiv:1906.00968), attached to this sentence in an earlier draft, is a library of *isotropic*, spherically-averaged matter power spectra and an empirical suppression-amplitude-vs-$f_{\rm gas}$ model – it does not document angular gas-map anisotropy and does not support this specific claim (verified against the full text).] What has not been measured is how much of the WL observable itself – the lensing convergence power spectrum $C_\ell^{\kappa\kappa}$ that stage-IV surveys use to marginalize over baryons – is actually carried by this discarded angular information, as opposed to the radial profile a spherical BCM already captures.

Answering this at the field level requires a generative model that (i) paints full hydrodynamic-quality baryon fields halo-by-halo, so that (ii) one can build a faithful, per-halo spherical counterfactual of the *same* halos under the *same* realization, and difference it against the full field. BIND (Lee, 2026a, Paper I) is exactly this model: a conditional flow-matching emulator that paints [$\mathrm{DM_{hydro}}$, Gas, Stars] maps onto DMO halo cutouts, conditioned on the CAMELS-IllustrisTNG 35-parameter cosmology+astrophysics vector. In this Letter we use the BIND lightcone pipeline to build, for the first time, a literal radial counterfactual of a full painted WL field and measure the fraction of the convergence-power suppression that a spherically symmetric baryonification model can never reproduce. We report this fraction as a function of scale and identify the feedback channels that control it, while being explicit that our fiducial control is a conservative, first field-level bound rather than a final number (§4).

2 Methods

2.1 Three fields, one set of halos

For every halo along the BIND lightcone we build three versions of the same patch from the same DMO particles and the same painted realization: a `dmo` field (no baryons), a `bind` field (the full BIND composite, i.e. the hydrodynamic-quality painted map), and a spherical control field that retains only the monopole of the `bind` correction. Convergence maps κ are then built by ray-tracing the same lightcone through all three fields, so that cosmic variance, projection/shape noise, and halo sampling cancel exactly in any difference between them.

2.2 The faithful radial-BCM-warp control

The spherical control is the crux of the measurement, and we use two versions of it here for a specific reason. An earlier implementation (`azimuthal_average_patch`, "mono") simply azimuthally averages the *correction* $\Delta\Sigma = \Sigma_{\rm hydro} - \Sigma_{\rm dmo}$ in each patch. This forces the correction itself to be circular, which is *not* what a real BCM does: an actual radial-push BCM (potential- or mass-based displacement, in the spirit of e.g. Dai et al. 2018) moves mass along $r \to r'(r)$, so a triaxial halo's corrected surface density remains triaxial even under a perfectly "spherical" baryon model – only the *displacement law* is radial, not the resulting map. The mono control therefore credits a real BCM with zero triaxiality-tracing ability, which inflates the apparent missed anisotropy by an amount set by the halo's intrinsic (DM) triaxiality (Osato et al., 2018).

Our primary, faithful control (`bcm_warp_patch`, "sph"; `mode="bcm_warp"` in `bind.inference.spherical`) instead pushes each DMO pixel radially so that $M_{\rm dmo}(<r) = M_{\rm hydro}(<r')$ is satisfied mass-exactly

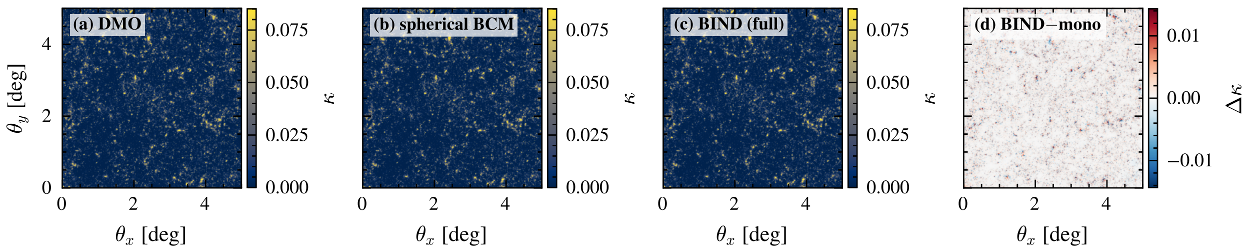

Figure 1: Convergence maps built from the *same* halos and the same lightcone realization at $z_s = 1$: DMO-only, the crude pure-spherical `mono` control (panel label "spherical BCM"), the full BIND composite, and their difference $\kappa_{\rm bind} - \kappa_{\rm mono}$ (rightmost panel). *This figure uses the earlier, superseded `mono` control (§2.2), not the faithful `bcm_warp` control* – no rendered `bcm_warp` map exists in our source material at the time of writing. The difference panel is therefore an *upper bound* on the pure angular signal: it also contains the triaxiality-tracing term $f_{\rm tri}$ that a faithful radial BCM would remove (§3.1), not only the genuine feedback anisotropy $f_{\rm aniso}$ – but it is still visually concentrated on massive, triaxial halos rather than spread uniformly across the field.

(CIC-interpolated) about the halo's own center. Because the push is purely radial, halo triaxiality *co-moves* with the mass flow exactly as it would under a real BCM; only genuine *angular* feedback structure – azimuthal non-uniformity in where gas is deposited or ejected, beyond what a radial push of a triaxial halo already produces – separates `sph` from `bind`. `bcm_warp_patch` itself was verified at the time of this control's introduction to conserve mass to 10^{-5}, track the hydro monopole, and preserve the DMO annular RMS, but that verification is qualitative only in our source material. [TODO: a quantitative L2/correlation validation of `bcm_warp_patch` specifically, of the kind reported below for `mono`, has not been computed in the sources available to us.] Separately – and this is a validation of the crude `mono` control, *not* of `bcm_warp_patch` – we stacked the monopole $c_0(r)$ of the BIND correction against the azimuthally-averaged `mono` correction over cluster-mass halos: the two agree to an L2 relative residual of 2.22% with correlation 0.99998, and the residual `bind` − `mono` is a pure $m \geq 2$ field, with its own monopole consistent with zero at the $\mathrm{RMS}|c_0|/\mathrm{RMS}|c_2| = 0.018$ level (i.e. the `mono` control leaves the radial profile intact to $\sim$2% while removing only the quadrupole-and-higher structure). For the single most massive halo in the sample ($\log_{10}(M_{200c}/\mathrm{M}_\odot h^{-1}) = 14.57$), the anisotropic component alone carries 97% of the correction's pixel RMS, illustrating how large the angular term can be for an individual triaxial cluster. This only shows that `mono` is a fair monopole-preserving counterpart to `bind`; per the argument above, it does *not* by itself validate `bcm_warp_patch`, since the `mono` residual conflates genuine feedback anisotropy with the triaxiality-tracing term $f_{\rm tri}$ that a faithful radial BCM would instead capture (§3.1). Figure 1 shows one realization of the resulting fields built with the `mono` control, the only spherical control for which a rendered map exists in our source material at the time of writing; the faithful `bcm_warp` control's field-level numbers instead appear in Figure 3 (§3.1).

2.3 Field statistic and the first-order mechanism

For any map statistic X we define the total baryonic effect $\Delta X_{\rm baryon} = X_{\rm bind} - X_{\rm dmo}$, the anisotropic component $\Delta X_{\rm aniso} = X_{\rm bind} - X_{\rm sph}$, and the anisotropic fraction $f_{\rm aniso}(X) = \Delta X_{\rm aniso}/\Delta X_{\rm baryon}$. For the convergence power spectrum this admits an exact decomposition,

$$\Delta P \equiv P_{\rm bind} - P_{\rm sph} = \underbrace{2\,\mathrm{Re}\big\langle \mathcal{F}[\kappa_{\rm dmo}]^*\,\mathcal{F}[\kappa_{\rm bind} - \kappa_{\rm sph}]\big\rangle}_{\text{cross, first order}} + \underbrace{\big\langle |\mathcal{F}[\delta_{\rm bind}]|^2\big\rangle - \big\langle |\mathcal{F}[\delta_{\rm sph}]|^2\big\rangle}_{\text{auto, second order}}, \qquad (1)$$

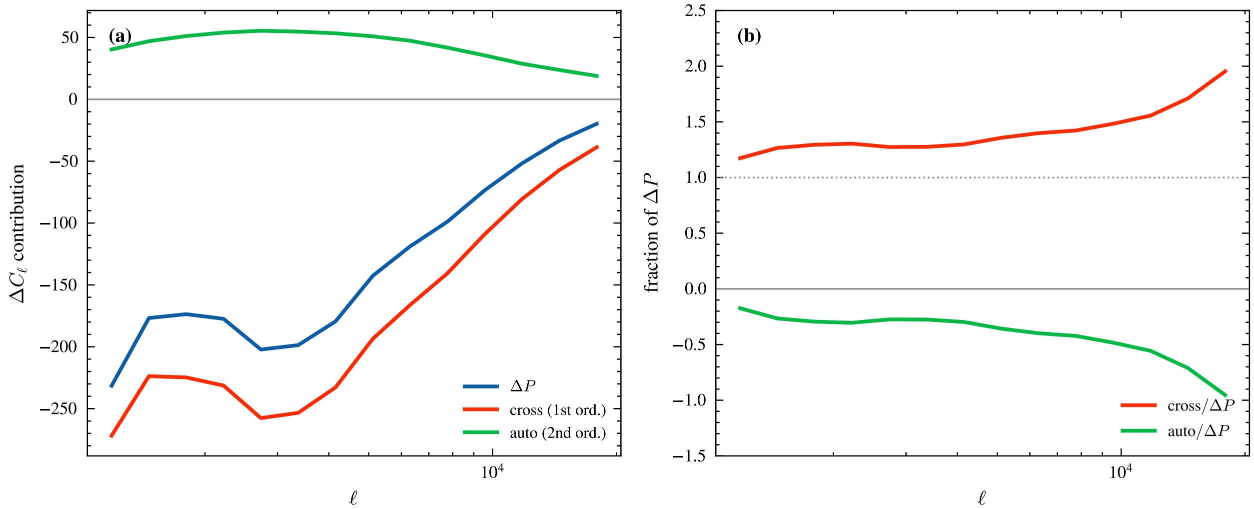

Figure 2: The exact cross/auto decomposition of $\Delta P = P_{\rm bind} - P_{\rm sph}$ (left: raw contributions; right: fraction of ΔP), showing that the cross (first-order) term dominates and has the correct sign to suppress power, while the auto (second-order) term is subdominant and of the wrong sign. This measurement uses the earlier, superseded circular/monopole control (§2.2); it motivates the first-order alignment mechanism qualitatively but has not yet been re-derived against the faithful `bcm_warp` control (see §4).

with $\delta \equiv \kappa - \kappa_{\rm dmo}$. The cross term is first order in the angular correction because it is linear in $(\kappa_{\rm bind} - \kappa_{\rm sph})$, weighted by the (large) DMO field itself; the auto term is quadratic in the (small) correction and can even have the "wrong" sign. Physically, the cross term dominates when the angular correction is spatially aligned with the halo's own DMO lensing signal – expected, since feedback-driven gas redistribution follows the halo potential, i.e. the same matter that lenses. On the fiducial lightcone (8 paired realizations, $z_{\rm s} = 1$, band $\ell \in [3 \times 10^3, 2 \times 10^4]$) this identity closes to machine precision and gives cross/$\Delta P \approx +1.39$ and auto/$\Delta P \approx -0.39$ (rising to $\approx 1.95/-0.95$ by $\ell \sim 1.8 \times 10^4$), confirming that the anisotropic suppression is carried at first order by alignment with the DMO field rather than by the correction's own auto-power. We flag explicitly that this decomposition (Figure 2) was measured with the earlier, mono spherical control, not the faithful `bcm_warp` control of §2.2; re-deriving it against the faithful control is listed as an open item in §4.

2.4 The 60 one-at-a-time feedback runs

To probe which feedback channels control the anisotropic signal we use the CAMELS-TNG one-at-a-time (OAT) design: each of the 30 cosmological/astrophysical parameters is varied to its minimum and maximum bound with all others held fixed, giving 60 `twobound` runs, 58 of which had completed spherical-BCM and BIND maps at the time of writing (2 pending a map re-run).[1] All OAT maps to date – and hence every feedback-dependence result in §3.3 – were built with the

[1] The 58/60 figure is the field-level design loaded by `wl_anisotropy_paper.ipynb`'s Field-4 cells (§3.3). A "57 completed single-knob OAT runs" count also appears twice in our source material for a nominally similar design: once in the separate per-halo-physics notebook (`ejection_anisotropy.ipynb`), and once in `wl_anisotropy_paper.ipynb`'s own Pillar-3 per-halo cell (its local `decode_oat()` loader), which is the cell §3.3 cites for the per-halo quadrupole ranking. The 57-vs-58/60 discrepancy is therefore not cleanly a cross-notebook mismatch – it is partly internal to `wl_anisotropy_paper.ipynb` itself – and is not reconciled in our source material; we flag it as an open bookkeeping item rather than a physical result.

earlier, superseded mono control; the faithful `bcm_warp` three-way comparison (§2.2, Figure 3) has so far only been run at the fiducial point.

2.5 Error model

Because all three fields share the same halos, DMO particles, and lightcone realization seeds, differences between them cancel cosmic variance to a degree not available to independent hydrodynamic re-simulations: feeding the null triplet $(\texttt{dmo}, \texttt{sph}, \texttt{sph})$ into the pipeline returns $f_{\rm aniso} = 0$ exactly (no manufactured signal), and the paired-realization band-power scatter is $25.8\times$ smaller than the scatter between unpaired realizations, confirming that seed-sharing suppresses variance by this factor and that the measured signal is not a pairing artifact. Errors are nonetheless bootstrapped over only $N_{\rm real} = 8$ paired plane-randomizations of a single $205\,{\rm Mpc}/h$ box, so they capture realization/projection scatter but not independent cosmic volumes, and are consequently optimistic (§4). For the stacked per-halo forecast (§4) we also use an empirically validated DMO-shape-axis misalignment dilution $\langle \cos 2\Delta\theta \rangle = 0.55$ (mean misalignment $\Delta\theta \approx 28^\circ$, Gaussian rms $\sigma \approx 26^\circ$), which differs from the commonly assumed 35° figure.

3 Results

3.1 The corrected, scale-dependent anisotropic fraction

Figure 3 shows the three-way comparison at the fiducial feedback point ($z_{\rm s} = 1$) using the faithful `bcm_warp` control. We define $f_{\rm aniso} = (C_b - C_s)/(C_b - C_d)$ (angular structure a spherical BCM misses), $f_{\rm tri} = (C_s - C_m)/(C_b - C_d)$ (triaxiality-tracing that a real BCM *does* capture, isolated by comparing the faithful control s to the crude mono control m), and $f_{\rm mono} = f_{\rm aniso} + f_{\rm tri} = (C_b - C_m)/(C_b - C_d)$ (the old, upper-bound estimate). The headline result of this Letter is that the corrected fraction *declines* with scale: $f_{\rm aniso} \approx 0.45$ at $\ell = 3 \times 10^3$, 0.24 at $\ell = 10^4$, and ≈ 0 by $\ell = 2 \times 10^4$, while the triaxiality-tracing term *rises* over the same range, $f_{\rm tri} \approx 0.08 \to 0.25 \to 0.61$. The sum $f_{\rm mono} \approx 0.5$ is roughly flat in ℓ – reproducing the earlier, cruder estimate almost exactly – which is the direct demonstration that the old number was dominated at small scales by triaxiality a real BCM already reproduces, not by missed feedback physics. In real space, both controls track BIND almost equally well overall ($r(\texttt{bind}, \texttt{sph}) \approx r(\texttt{bind}, \texttt{mono}) \approx 0.987$–$0.988$, and the faithful control's real-space residual from BIND is statistically indistinguishable from the crude one's – both $\approx 16\%$ of the total κ RMS globally, i.e. the faithful control only "barely beats" `mono` overall), with the faithful control's advantage concentrated specifically at high ℓ where $f_{\rm tri}$ is largest.

3.2 Comparison with the earlier estimate

The number originally reported for this measurement, before the control was corrected, is worth recording for provenance since it still appears in the underlying analysis notebook's own narrative. Using the crude mono control, the band-averaged small-scale suppression deficit gives $f_{\rm aniso}(C_\ell) \approx 41\% \to 48\% \to 55\%$ at $\ell = 5000 \to 10^4 \to 2 \times 10^4$ – *increasing* with scale, the opposite trend to the corrected measurement above – with a band average over $\ell \in [3000, 20000]$ of 48% (bootstrap 16–84% interval 47–48%) at a paired significance of $z = 52.3\sigma$ (optimistic, shared box). As shown in §3.1, this number is now understood to be the sum of the genuine angular-structure fraction and the triaxiality fraction that a real BCM already captures; we do not use it as the Letter's headline result, and present it here only to document the correction.

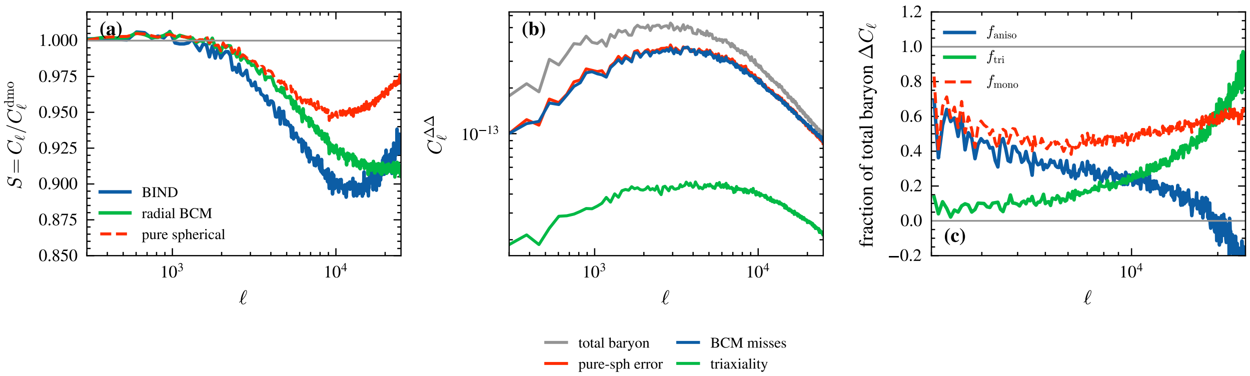

Figure 3: The corrected, faithful three-way comparison at the fiducial feedback point ($z_{\rm s} = 1$). *Left:* suppression $S(\ell) = C_\ell/C_\ell^{\rm dmo}$ for the full BIND field, the faithful radial-BCM-warp control, and the crude pure-spherical (monopole) control. BIND suppresses power well below either control at $\ell \gtrsim$ few$\times10^3$, and the radial-BCM-warp control tracks BIND substantially more closely than the pure-spherical control does, by construction. *Center:* residual-field power for the total baryonic effect, the pure-spherical error, the faithful-BCM miss, and the triaxiality term. *Right:* the decomposition fractions $f_{\rm aniso}$, $f_{\rm tri}$, and $f_{\rm mono} = f_{\rm aniso} + f_{\rm tri}$ vs. ℓ – the corrected, missed-angular-structure fraction $f_{\rm aniso}$ declines with scale while the triaxiality-tracing fraction $f_{\rm tri}$ rises, so their sum reproduces the old, scale-flat estimate almost exactly. This is the headline figure of the Letter.

3.3 Which feedback channels drive the (upper-bound) anisotropic signal

We rank the 30 OAT feedback parameters by the band-averaged ($\ell \in [2000, 20000]$) response of the anisotropic suppression $A = 100 \times (S_{\rm sph} - S_{\rm bind})$ to each parameter's min/max swing, using the fiducial-realization scatter ($\sigma = 0.161\%$) to set a 2σ significance floor of $|dA/du| > 0.456\%$; 22 of the 30 parameters move A above this floor. The strongest responders are, in order, BlackHoleRadiativeEfficiency ($dA/du = -9.90\%$), QuasarThreshold ($+6.21\%$), IMFslope ($+5.53\%$), SeedBlackHoleMass ($+4.02\%$), VariableWindVelFactor ($+3.91\%$), BlackHoleAccretionFactor ($+3.36\%$), BlackHoleEddingtonFactor ($+2.96\%$), and WindEnergyIn1e51erg (-2.93%). A complementary per-halo response of the group-scale quadrupole moment ranks VariableWindVelFactor, BlackHoleRadiativeEfficiency, WindEnergyIn1e51erg, WindFreeTravelDensFac, and QuasarThreshold among the top drivers as well, so the two independent rankings agree qualitatively: AGN kinetic-mode (radiative efficiency, quasar threshold, seed/accretion physics) and stellar-wind parameters dominate the anisotropic control, consistent with the ejection-orientation physics of §3.4. As emphasized in §2.4, these rankings are built entirely from the superseded mono control – no OAT sweep yet exists for the faithful `bcm_warp` control – so we present them as the drivers of the *upper-bound* angular signal, not of the corrected, scale-dependent $f_{\rm aniso}$ of §3.1.

3.4 Where the ejection is anisotropic, and why

The per-halo, DM-shape-axis-aligned stacks in Figure 4 show that the two probes most sensitive to feedback anisotropy respond very differently with radius. The WL convergence residual is only mildly anisotropic, $q_2(\kappa - \kappa_{\rm dmo}) \approx 3$–$5\%$ of the monopole and roughly flat to declining with radius, whereas the tSZ pressure quadrupole is strongly anisotropic in the outskirts, rising from $\sim$5–12% near the halo center to $\sim$35–43% at $r \approx 2.9\, r_{200c}$. The fiducial ejection quadrupole is anti-aligned with the halo's DM major axis, $\langle \cos 2(\theta_\Delta - \theta_{\rm DM}) \rangle = -0.773$, i.e. a bipolar signature in which baryons are preferentially redistributed along the halo's *minor* axis – consistent with collimated AGN-jet-

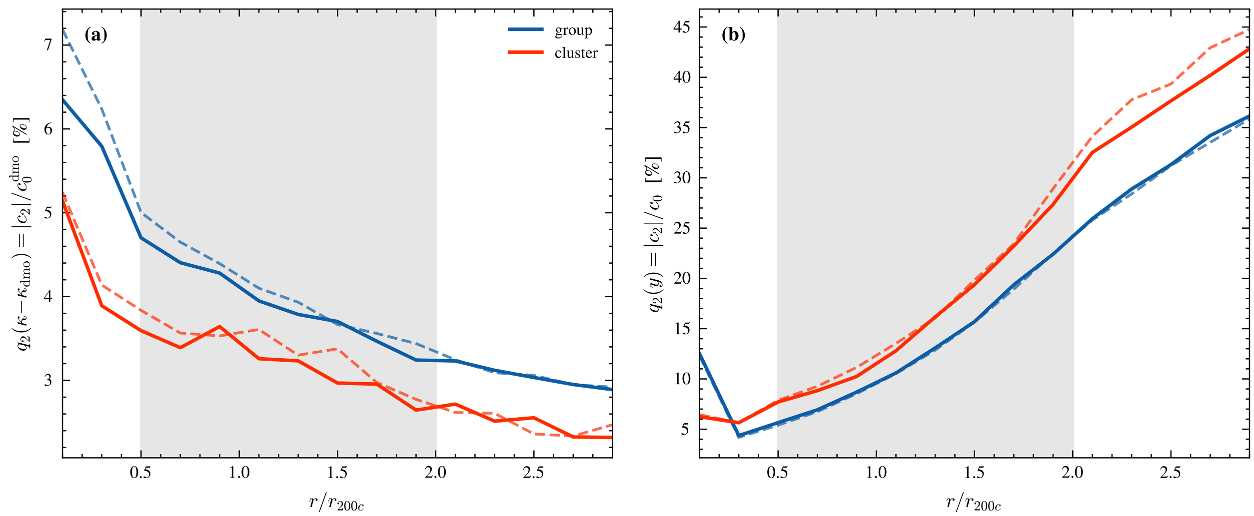

Figure 4: Radial profiles of the aligned quadrupole fraction $q_2 = |c_2|/c_0$ (or $/c_0^{\rm dmo}$ for κ) for the WL convergence residual (left) and the tSZ pressure (right), for group- and cluster-mass bins, comparing the fiducial BIND lightcone (solid) to TNG truth (dashed). Color encodes mass bin (group/cluster); only linestyle distinguishes BIND from TNG truth in this figure – unlike Figures 1-3, color here is not the bind/truth semantic. The WL residual is mildly anisotropic (3–5%) and roughly flat with radius, while the tSZ pressure quadrupole grows strongly in the outskirts ($\sim$35–43% at $r \sim 2.9\, r_{200c}$); the shaded band marks the annulus used for the ejection-orientation and environment tests. Weak-lensing κ is the cleaner, environment-insensitive probe of feedback-driven anisotropy.

like ejection rather than passive tracing of the DM shape. An environment jackknife further shows that the outer tSZ quadrupole is substantially contaminated by large-scale structure: halos in filament environments carry $\approx 52\%$ of the signal at $2.7\, r_{200c}$ versus $\approx 29\%$ in void environments, so the outer-y anisotropy is not purely an AGN-orientation effect. The WL convergence quadrupole, by contrast, shows no comparable environment dependence in our source material and is the cleaner feedback-only probe of the two.

4 Discussion

4.1 Implications for BCM-based baryon marginalization

A spherically symmetric BCM has, by construction, zero quadrupole and therefore zero cross term in the exact identity of §2.3: it can only match the true (anisotropic) $P(\ell)$ by over-ejecting gas radially to compensate for the missing angular channel. Concretely, under the (superseded) mono control the fiducial suppression reaches $S_{\rm sph} \approx 0.93$ at $k \sim 17\, h/{\rm Mpc}$ versus $S_{\rm bind} \approx 0.86$ for the same halos, so a spherical model tuned to match 0.86 in projection must inflate its radial ejection well beyond the level a real, mass-matched profile would require. This biases the recovered gas profile even when the two-point function is fit correctly, which has three testable consequences: (i) it breaks multi-probe consistency, since the same over-ejected profile that fits $C_\ell^{\kappa\kappa}$ will generically be wrong for tSZ y, X-ray $f_{\rm gas}$, and kSZ/τ observations of the same halos; (ii) it plausibly explains why spherical BCMs reproduce hydrodynamic non-Gaussian statistics only up to a limited peak height, since our own earlier work found BCM peak counts diverge from hydrodynamic simulations above

$\nu \gtrsim 4$ (Lee et al., 2023) – the missing ν tail is a natural consequence of a missing quadrupole; and (iii) because the cross term depends on the alignment between feedback geometry and halo shape, it need not transfer cleanly across feedback models or cosmologies, which a purely radial calibration cannot capture. These arguments motivate an explicit, calibrated angular ($m = 2$) correction term $c_2(r)$ alongside the radial $c_0(r)$ profile already used in BCM fits, of the kind BIND can supply per halo. We note that this is exactly the gap that data-driven nuisance parametrizations are positioned to close: our companion analysis (Lee, 2026b, Paper II) shows that a small number ($N = 2$) of data-driven baryon templates is sufficient to remove the TNG baryonic bias from a stage-IV cosmic-shear analysis, whereas analytic, spherically symmetric BCMs have no free parameter that can absorb an angular term by construction; the anisotropic fraction measured here is a plausible structural reason such templates outperform analytic BCMs at fixed parameter count.

4.2 Detectability

A stacked, DM-shape-aligned tangential-shear quadrupole is in principle directly observable, subject to the $\langle \cos 2\Delta\theta \rangle = 0.55$ alignment dilution of §2.5. For the cluster-mass bin, the forecast signal-to-noise at $N_{\rm cl} = 10^4$ reaches 1.5 for LSST Y10, 1.5 for Euclid, and 2.3 for Roman HLS (rising from 0.5/0.5/0.7 at $N_{\rm cl} = 10^3$ and 0.8/0.8/1.3 at $N_{\rm cl} = 3 \times 10^3$); the group-mass sample forecast is lower still across the sampled range. These forecasts assume self-similar cluster profiles placed at $z_l = 0.3$ and measured at $z \sim 0$ from CAMELS-SB35 training data, which caps cluster mass at $M_{200c} \lesssim 5 \times 10^{14}\,{\rm M}_\odot\,h^{-1}$ and imposes a $\sim 50\,{\rm kpc}/h$ resolution floor from the 128^2-pixel patches used to train BIND, so they should be read as indicative rather than survey-ready. A stacked-quadrupole detection at $N_{\rm cl} \sim 10^4$ therefore looks achievable only marginally, at $\lesssim 2\sigma$ with current alignment-dilution assumptions, in the next generation of cluster lensing surveys.

4.3 Caveats

This is a preliminary, first field-level measurement, and several caveats bound how it should be read. First, the corrected, declining anisotropic fraction of §3.1 is itself only a partial revision: it has so far been measured only at the fiducial feedback point, so §3.3's ranking of which parameters control the anisotropy still uses the earlier, upper-bound control, and no feedback-dependence of the corrected $f_{\rm aniso}$ yet exists. Second, the first-order cross/auto mechanism of §2.3, which motivates *why* the anisotropy reaches the power spectrum at all, was itself derived against the superseded control; whether the cross term still dominates when computed against the faithful `bcm_warp` control has not been re-verified, and we flag this as an explicitly open item rather than assume it carries over. Third, the reported fraction rests on a single fiducial-feedback, fixed-cosmology TNG lightcone; we do not yet know whether $f_{\rm aniso}(\ell)$ itself depends on feedback strength or on cosmology. Fourth, our uncertainties are bootstrapped over $N_{\rm real} = 8$ paired plane-randomizations of one $205\,{\rm Mpc}/h$ box and therefore capture realization/projection scatter, not independent cosmic volumes; they should be read as optimistic (§2.5). Finally, the non-Gaussian (peak/PDF) significance estimates that motivated §3.4's discussion used per-field marginal errors rather than the paired covariance, which is conservative and should understate the true significance of any paired anisotropic peak-count signal; quantifying that signal directly against the corrected control is left to future work.

5 Conclusions

Using BIND's per-halo generative painting to build, for the first time, a mass-exact and triaxiality-faithful spherical counterfactual of a full weak-lensing field, we measure the field-level fraction of the baryonic convergence-power suppression that no spherically symmetric baryonification model can reproduce. The corrected fraction is scale-dependent, $f_{\mathrm{aniso}} \approx 0.45 \to 0.24 \to 0$ from $\ell = 3\times10^3$ to 2×10^4, substantially smaller at small scales than an earlier, cruder circular-control estimate of $f_{\mathrm{mono}} \approx 0.5$ (roughly flat in ℓ), which we show is inflated at small scales by halo triaxiality that a real radial BCM already reproduces. The 60-parameter one-at-a-time TNG feedback sweep points consistently to AGN kinetic-mode and stellar-wind parameters as the dominant controls of this (upper-bound) angular signal, and the fiducial ejection is bipolar and anti-aligned with the halo's DM major axis. Even where the corrected fraction is small, its non-zero existence implies that fitting the two-point function with a spherical BCM requires an unphysical radial over-ejection of gas, biasing the recovered profile in a way that plausibly explains known BCM failures on non-Gaussian statistics and that a purely radial, analytic marginalization cannot self-correct. We conclude that stage-IV baryon marginalization pipelines should budget for an explicit angular correction term, calibrated the way BIND calibrates it here, rather than assume a radial-only model is structurally sufficient. This measurement is preliminary in the specific ways enumerated in §4.3, and we plan to extend the faithful control across the full OAT feedback grid and re-derive the first-order mechanism against it in follow-up work.

Acknowledgments

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